

Practice Quiz: Cognition -- The Models Inside Your Head

Part A: Multiple Choice

1. A **generative model** in Active Inference is best described as: A) A perfect copy of reality stored in your brain B) Your brain's simplified, compressed theory of how causes in the world produce the effects you observe C) A textbook you have memorized D) A model that only works for physical objects

Answer: B. The generative model is a theory of causal structure that generates predictions, not a perfect replica of reality.

1. In Active Inference, a **belief** is: A) An opinion you consciously argue for B) A parameter in your generative model that shapes predictions, often operating below awareness C) Something you only hold about religion or politics D) A feeling with no connection to prediction

Answer: B. Many beliefs are implicit (like trusting that floors support your weight) and only noticed when they produce prediction errors.

1. Confirmation bias -- seeking evidence that supports existing beliefs -- is best understood as: A) A character flaw showing laziness B) A feature of an efficient generative model that assigns high prior probability to its own predictions C) A rare disorder D) Something only unintelligent people do

Answer: B. Confirmation bias is how an efficient inference engine works under limited time and energy, though it can cause persistent errors.

1. Active Inference suggests that **emotions** are: A) Irrational disruptions that interfere with thinking B) Signals about the state of your generative model -- indicators of prediction error, uncertainty, and confidence C) Completely random chemical reactions D) Only useful for artistic expression

Answer: B. Anxiety signals high expected prediction error; sadness signals a significant model update; joy signals low prediction error in valued domains.

1. Your generative model is **hierarchical**, meaning: A) It only has one level B) Surface-level predictions are nested within mid-level models, which are nested within deep models -- with deeper levels changing more slowly C) Higher levels are always more accurate D) It can only process visual information

Answer: B. The model has layers of depth, and updating deep beliefs ripples through many dependent predictions.

1. People resist changing deeply held beliefs even with clear evidence because: A) They are stubborn by nature B) Deep model revisions propagate through many dependent predictions, making the free energy cost enormous C) Evidence is always unreliable D) Beliefs cannot change after age 18

Answer: B. The deeper the belief in the hierarchy, the more predictions depend on it, and the costlier the revision.

1. **Rumination** (going over the same worrying thoughts repeatedly) can be understood as: A) A healthy thinking strategy B) A generative model stuck in a local minimum where anxiety-filtered evidence confirms its own threat predictions C) A sign of high intelligence D) Something that only happens to adults

Answer: B. The model minimizes free energy locally (predictions match anxiety-filtered evidence) but fails globally.

1. When you feel anxious before a big test, Active Inference explains this as: A) Your body trying to make you fail B) Your generative model signaling high expected prediction error and uncertainty about what will happen C) Proof that you did not study enough D) A random emotional response with no meaning

Answer: B. Anxiety is an informative readout from your inference machinery about expected future surprises.

1. The balance between **perceptual inference** (changing beliefs to fit evidence) and **active inference** (changing the world to fit beliefs) is central to healthy cognition because: A)

You should always change your beliefs B) Excessive belief-updating causes instability while excessive world-changing causes inflexibility C) Only one strategy ever works D) The balance does not matter

Answer: B. Healthy cognition requires flexibility to update beliefs AND agency to shape the world.

1. When someone resolves cognitive dissonance by saying "they deserved it" rather than updating their self-model, they are: A) Being honest B) Reinterpreting evidence to preserve existing beliefs rather than performing genuine model updating C) Always making the right choice D) Eliminating all prediction error permanently

Answer: B. Both strategies minimize free energy, but only genuine model updating leads to a more accurate generative model.

Part B: Short Answer

1. Identify a belief you hold that operates below conscious awareness -- something so basic you never question it. What would happen to your daily predictions if it turned out to be wrong?
2. Explain how understanding emotions as "model diagnostics" might change how you respond to feelings of anxiety before a presentation or test.
3. Give an example from your life where changing a deep belief (about yourself, a relationship, or the future) was much harder than changing a surface-level opinion. Why does Active Inference predict this difficulty?
4. Describe a situation where someone you know seemed trapped in a rumination loop. Using Active Inference, explain what might help break the cycle.
5. How might cognitive biases like anchoring or the availability heuristic actually be useful in everyday decision-making, even though they sometimes lead to errors?